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Learning Or Modelling? An Analysis Of Single Image Segmentation Based On Scribble Information

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Motivation

- **Image Segmentation**: division into "meaningful" non-overlapping regions
- Segmentation networks: trained on thousands of examples and are therefore able to learn common shapes of objects in their training data
- **Limitation of segmentation networks**: only predict those semantic labels (and objects) present in the training database limits the applicability of such models
- **What is a meaningful region?** The answer, of course, is highly subjective and depends on the specific intended application.

Figure 1. The different ways of annotating the image on the left.

This is the reason why we believe that image segmentation on only one image based on scribbles, i.e., the prediction of regions in a single image that have previously been marked by a few strokes, so-called *scribbles*, by a user, remains highly relevant, e.g. for image editing software.

Model- and Learning-based Segmentation Methods

- **Objective**: Image segmentation solely given each pixel by a vector $x \in \mathbb{R}^n$:

$$\min_{\theta} CE(G(x,\theta),\ sc) \tag{1}$$

  using cross entropy ($\boldsymbol{CE}$) between the scribbles $\boldsymbol{sc}$ and the output of the network $\boldsymbol{G(x,\theta)}$.
- network $G(x,\theta)$ suggesting a class label for each pixel based on
  - ... solely given the RGB color x for $n=3$
  - ... given the color concatenated in the channel dimension with its x- and y-coordinates ($n=5$)
  - ... given a concatenation of the color and prior semantic information ($n=5$)
- training only on pixels that are covered by scribbles
- possible prior information: Use soft semantic features of Aksoy et al. [1], that have been successfully used in soft semantic segmentation without scribble information.
- **Spatially varying color distributions (SVCD) [2]** are used to model smoothly changing histograms in space to approach the scribble-based segmentation problem.
- **Pixelwise Networks (PWNs)** $G(x,\theta)$ are trained to mimic the behavior of color histogram-based approaches
- **Convolutional neural networks (CNNs)** with larger receptive fields: Using larger convolution kernels and increasing the depth of the networks allows us to provide the network with more and more non-local information.
- Here our objective is

$$\min_{\theta} CE(G(x,\theta),\ sc) \text{ with } x \in \mathbb{R}^{1\times n\times w\times h} \tag{2}$$

  with w, h being the image width and height and only the scribbled pixels contributing to the input gradient.
- **U-Net Architecture**, a convolutional architecture with a receptive field that spans large portions of the image while still being able to preserve fine details. The objective is similar to (1).

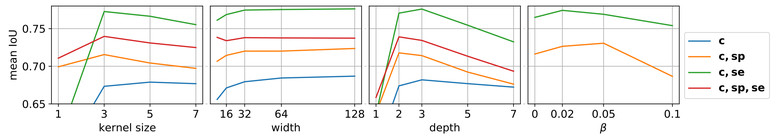

Figure 2. Study of the impact of the convolutional neural network from left to right: kernel size, width , and depth on the segmentation accuracy for different types of network input: **c**olor image, **sp**acial pixel position and **se**mantic information. The most right image shows the regularization of the spatial impact on the segmentation as defined in (3) (orange plot), as well as the corresponding penalty w.r.t. variations of the color channels (green plot).

Regularization

- **Problem**: inclusion of spatial information often dominates the results of the CNNs → objects close to the scribble are incorrectly segmented.
- **Regularization** term similar to the idea of *double backpropagation* from [3], to attenuate the influence of the spatial information:

$$\min_{\theta} L + 10^{-7}\beta||\nabla_{x_s} L||_1,\ L = CE(G(x_c, x_s, \theta), sc). \tag{3}$$

- The network loss L is computed by the cross entropy ($\boldsymbol{CE}$) between the scribbles $\boldsymbol{sc}$ and the output of the network $\boldsymbol{G}$, using the color information $\boldsymbol{x_c}$ and the spatial information $\boldsymbol{x_s}$.
- Regularization with semantic input data: similar to (3), but using the gradient with respect to the color input x_c instead of the spatial coordinates x_s.

$\beta = 0$ $\beta = 0.05$ $\beta = 0.1$

Figure 3. Segmentation with color and weighted spatial information via double backpropagation as network input.

Figure 4. Single image segmentation based on scribbles that rely on image color and on optional spatial or semantic information: The upper row shows the segmentation by SVCD, the second row shows the segmentation by a CNN.

Evaluation

In both settings (with and without prior information), we compare model- and learning-based segmentation methods:

	c	c & sp	c & se	c & sp & se
SVCD	-	0.775	-	0.845
PWN	0.607	0.721	0.689	0.731
CNN	0.673	0.715	0.772	0.739
CNN + reg.	-	0.731	0.774	-
U-Net	0.654	0.642	0.658	0.648

Table 1. Summary of the best mean IoU each of the pure single image segmentation methods could attain for different types of network input: **c**olor image, **sp**acial pixel position and **se**mantic information.

- model-based SVCD approach outperformed all learning-based approaches
- learning based approaches are improved by inclusion of the spatial coordinates as inputs to the neural network (except for U-Net)
- including our proposed regularization to avoid an overfitting to the spatial information only, gave the best result among the learning based approaches
- the combination of semantic and color information increases the performance significantly (except for U-Net)

References

[1] Y. Aksoy, T.-H. Oh, S. Paris, M. Pollefeys, and W. Matusik, "Semantic soft segmentation," *ACM Transactions on Graphics*, vol. 37, no. 4, 2018.

[2] C. Nieuwenhuis and D. Cremers, "Spatially varying color distributions for interactive multilabel segmentation," *TPAMI*, vol. 35, no. 5, 2012.

[3] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, 1998.